

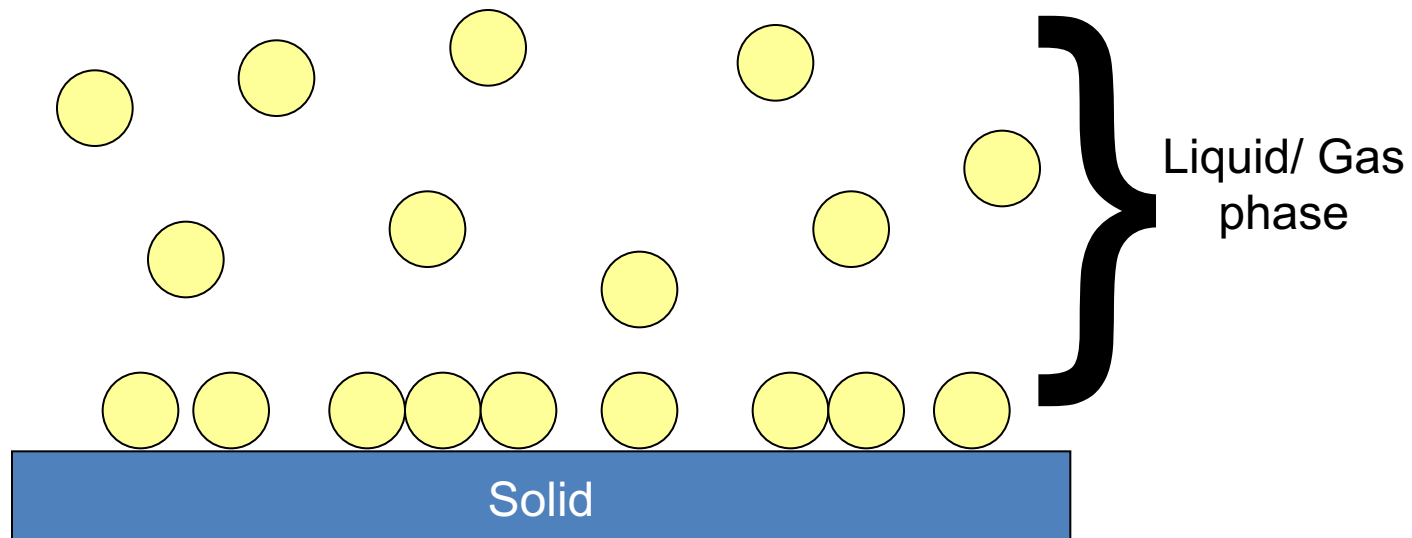
Protein Adsorption

What is a surface/ an interface?

The outermost region of a material that is chemically and/or energetically different from rest of the bulk by virtue of being located at a boundary.

What is adsorption?

Adsorption is formally defined as the partitioning of a chemical species between a bulk phase and an interface



adsorption vs. absorption

Adsorption vs. Absorption: Surface vs. Bulk phenomena

Both phenomena are relevant to biomaterials

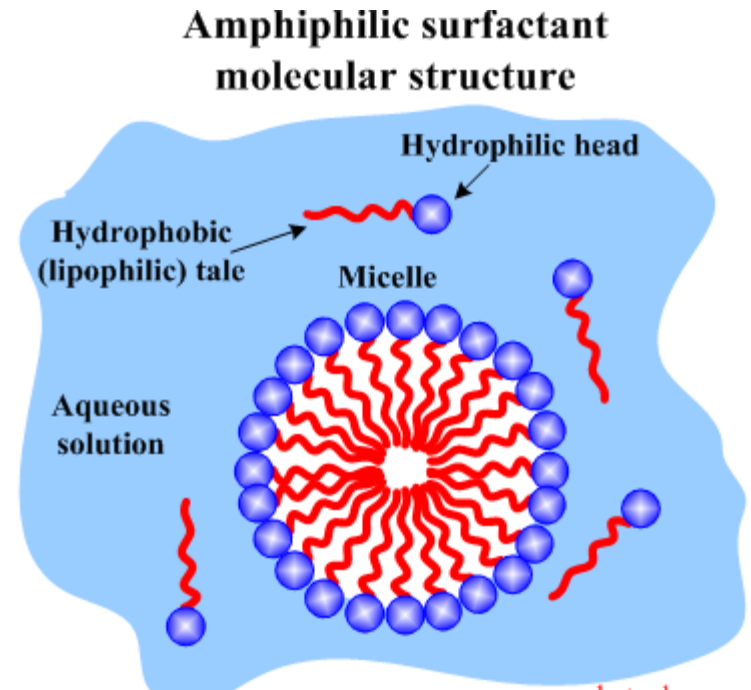
For example,

- When a dry crosslinked polymer network is put in water, it will absorb water to form a hydrogel
- Proteins adsorb to a biomaterial surface

What is a surfactant?

Surfactants are usually compounds that are amphiphilic: i.e., contain both hydrophobic groups (their *tails*) and hydrophilic groups (their *heads*).

Common examples:
Detergents and soaps



Why protein adsorb?

Proteins are *weak* surfactants (relative to synthetic detergents)

Because certain regions of a protein are richer in polar and charged amino acid residues (hydrophilic) whereas other regions are richer in non-polar residues (hydrophobic). There is a relative difference in hydrophobicity but not a large difference. So, they can easily change their structure to adapt to the surface encountered

Protein adsorption to biomaterial

Most bodily fluids (in vivo) and cell culture medium (in vitro) contain several soluble proteins

For example, blood contains > 400 different proteins at different concentrations

The sequence of events when a material is placed in a biological environment:

1. Interaction with water (wetting)
2. Interaction with proteins
3. Interaction with cells

CONSEQUENCE: *The adsorbed protein layer mediates the biological response to a biomaterial*

Protein may denature on adsorption

Protein structures are not highly stable.

Change in environment (heat and chemical agents) can cause a change in structure (denaturation).

When proteins adsorb (interaction with a solid) there is change in the surrounding chemical environment, thus potential for change in protein conformation (denaturation)

Some protein structures are more stable than others, so wide spectrum of denaturation depending on the protein and the surface with which it is interacting

How proteins denature on adsorption?

Denaturation will depend on biomaterial surface chemistry and water-wettability.

At a hydrophilic biomaterial surface (rich in charged and/or groups)-Hydrophilic amino acid rich regions of the protein will preferentially interact with the surface

So low denaturation as hydrophilic domains are on the outside

At a hydrophobic biomaterial surface (rich in non-polar groups)-Hydrophobic amino acid rich regions will preferentially interact with the surface

In water-soluble globular proteins hydrophobic amino acid residue-rich regions are in the protein core. These regions may prefer to interact with the hydrophobic biomaterial surface.

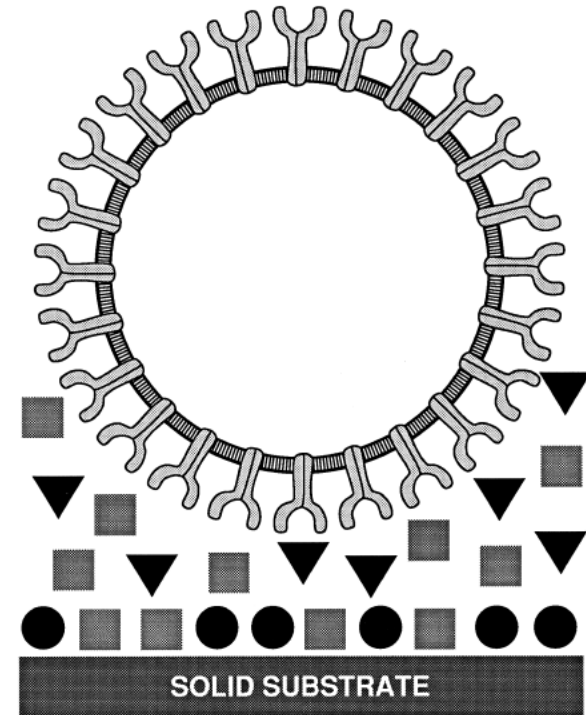
Why is protein adsorption important?

The adsorbed protein layer mediates the biological response to a biomaterial

Cell membrane has receptor proteins including “integrins” which bind to extracellular matrix (ECM) proteins in vivo

Cells attach to/ recognize biomaterials through binding of integrins to the adsorbed proteins.

- Immune response/ blood platelets
- Cell attachment/ growth for bio-integration of implants
- Cell attachment/ growth into 3D tissue scaffolds



Pre-adsorbing proteins to surface

Pre-adsorbing of cell adhesive proteins (such as fibronectin, fibrinogen, vitronectin) by incubating of the biomaterial in protein solution increases attachment of cells to the biomaterial surface.

Whereas, pre-adsorbing with non-cell adhesive proteins (such as albumin) decreases cell attachment to the biomaterial.

Often used to define the material signalling in-vivo so as to prevent the non-specific protein adsorption

Kinetics of protein adsorption

Very rapid initial rate of protein adsorption; rate limited by diffusion rates

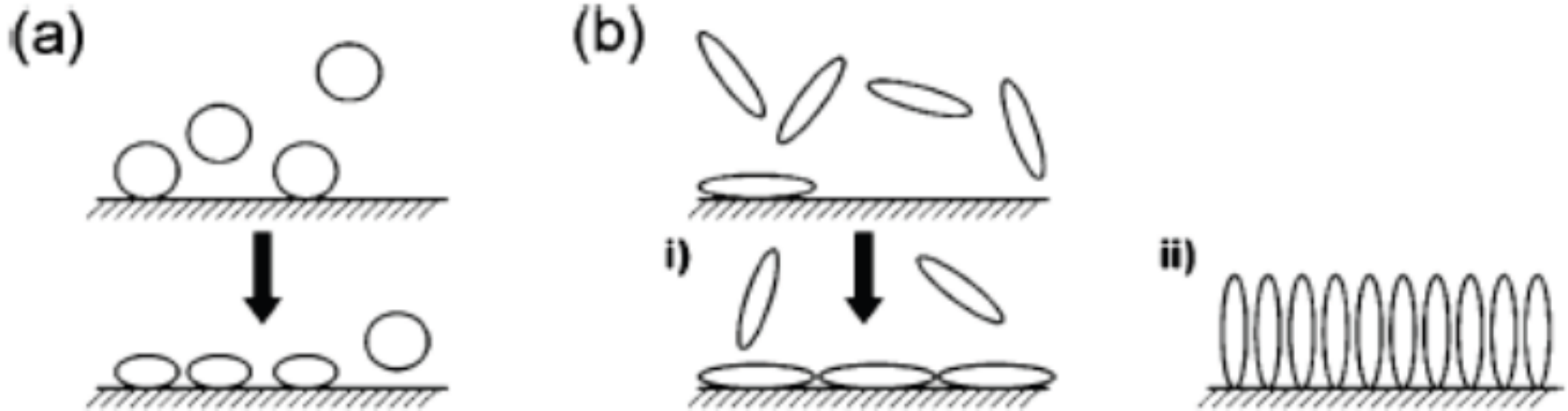
Then slower phase, more difficult for the arriving proteins to find and fit into an empty spot on the surface.

“Hard” and “Soft” proteins

“Hard proteins”: molecules with high internal stability for example; α -chymotrypsin, ribonuclease, lysozyme and β -lactoglobulin (β -Lg.)

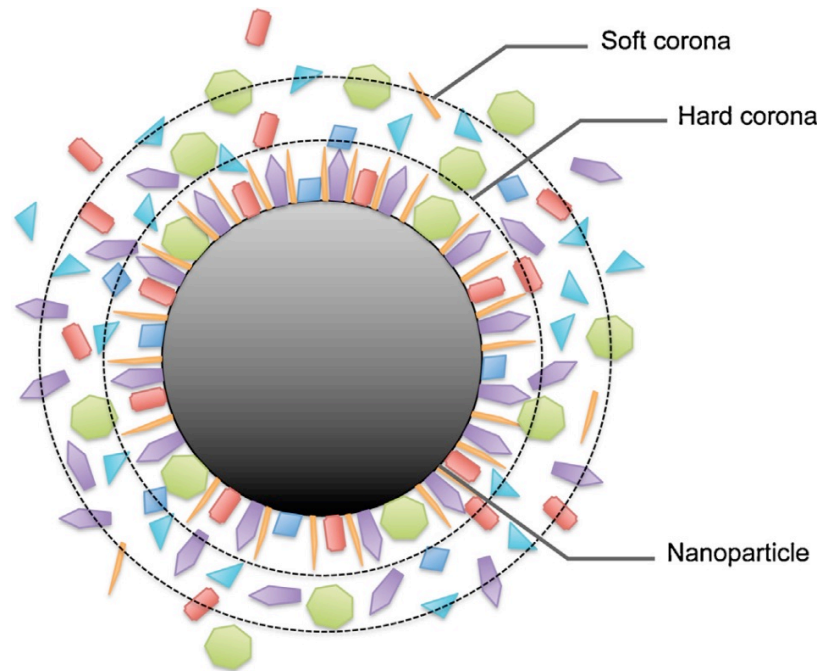
“Soft proteins”: have a low internal stability for example; α -lactoalbumin, β -casein, haemoglobin, catalase and phytase.

Change in protein orientation



Multilayer model

A 3D protein-rich “interphase” region is formed by the adsorbed proteins



Competitive adsorption

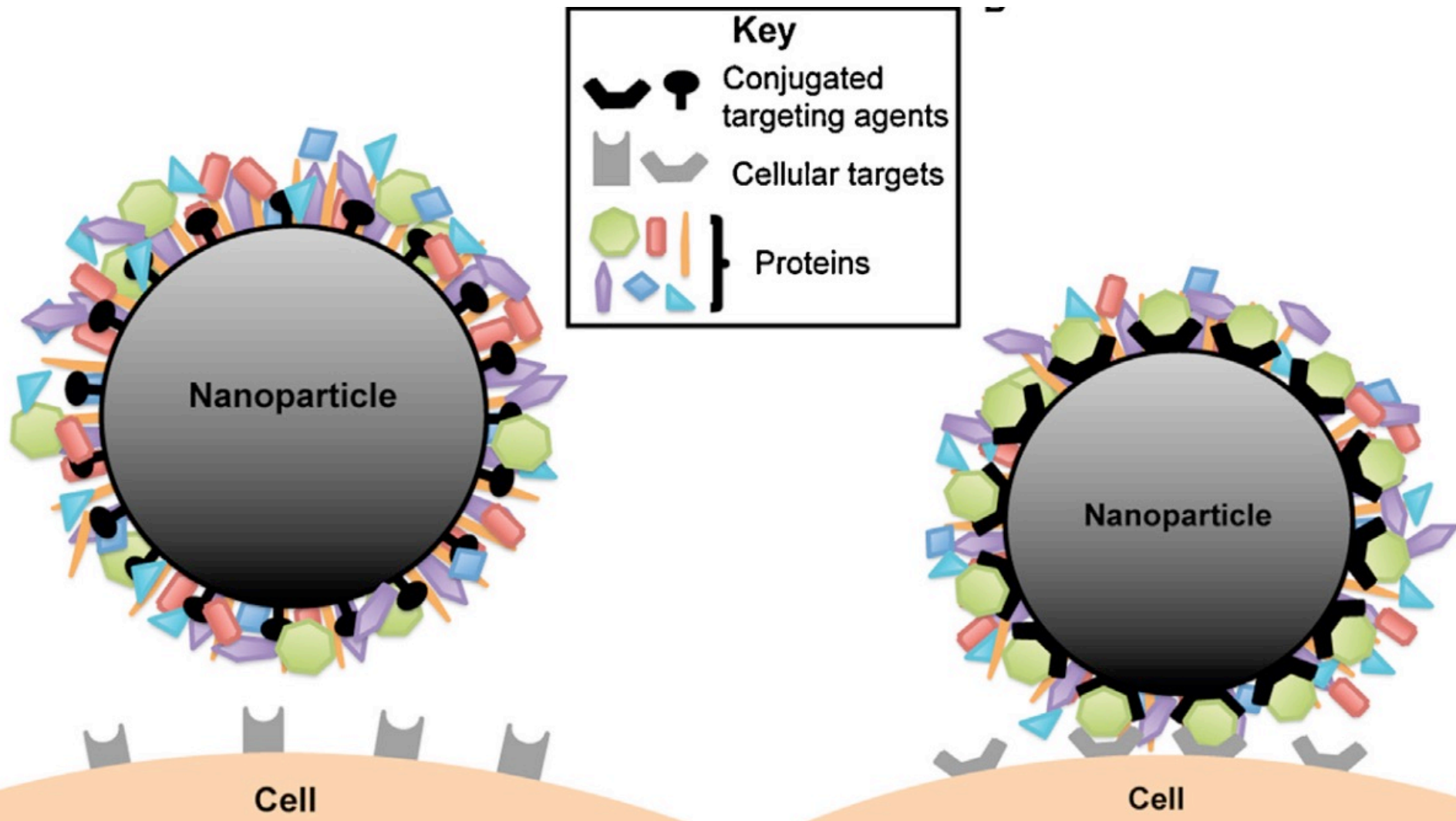
The equilibrium adsorbed protein layer composition determined by total and relative protein concentrations. Very complex phenomena and poorly understood.

Kinetics is also important. The composition of the adsorbed layer may change over time.

Adsorbed protein layer composition is dynamic

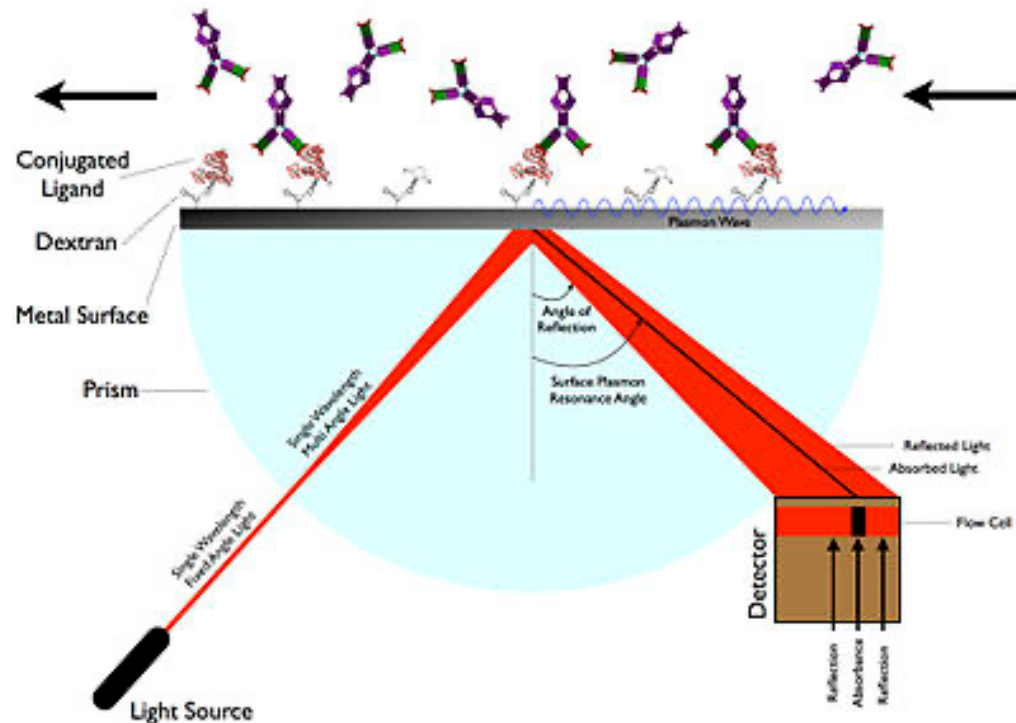
Initially-adsorbed proteins (kinetically favored) displaced by other proteins over time (minutes-to-hours-to-days) determined by thermodynamics

This is known as as Vroman effect



Protein adsorption quantification

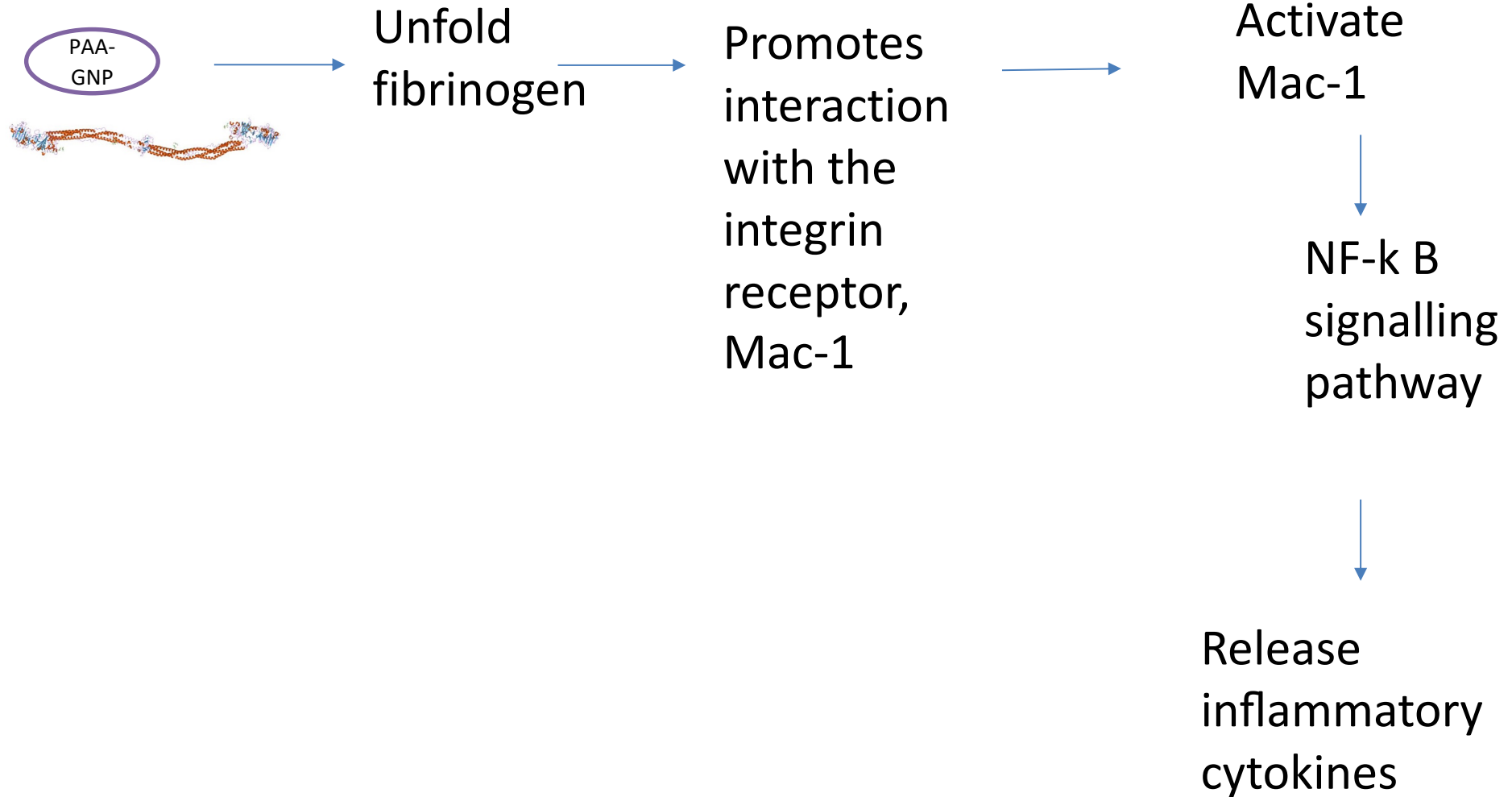
- Incubate, centrifuge followed by SDS PAGE gel
- Western blotting
- Surface Plasmon Resonance is widely used: Sensitive to dielectric of the surface which changes with adsorbed molecules

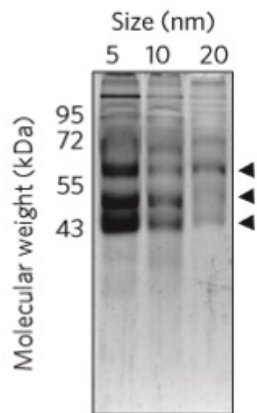
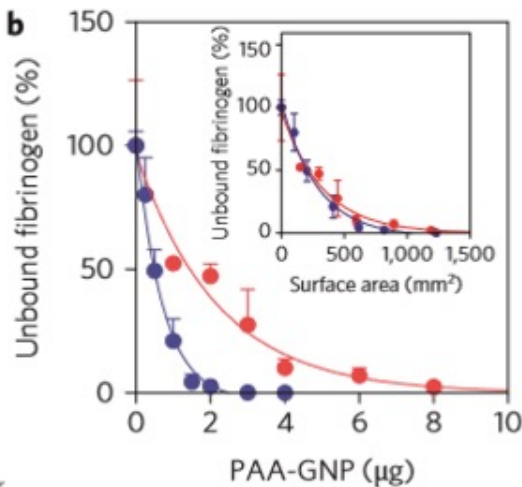
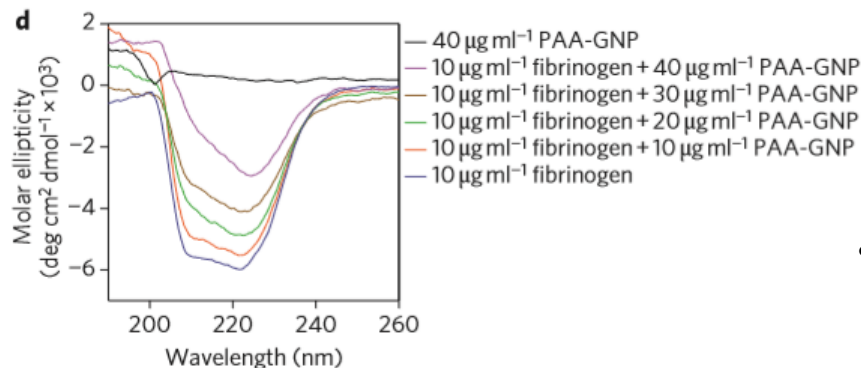


Nanoparticle-induced unfolding of fibrinogen promotes Mac-1 receptor activation and inflammation

Zhou J. Deng¹, Mingtao Liang^{2,3}, Michael Monteiro⁴, Istvan Toth^{2,3} and Rodney F. Minchin^{1*}

“Nanoparticles bound with proteins result in physiological and pathological changes”

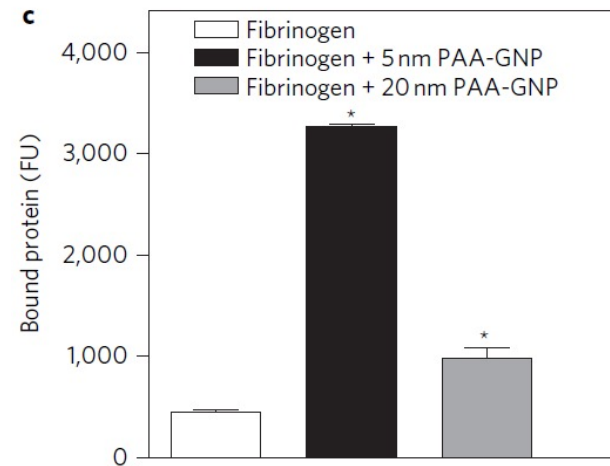
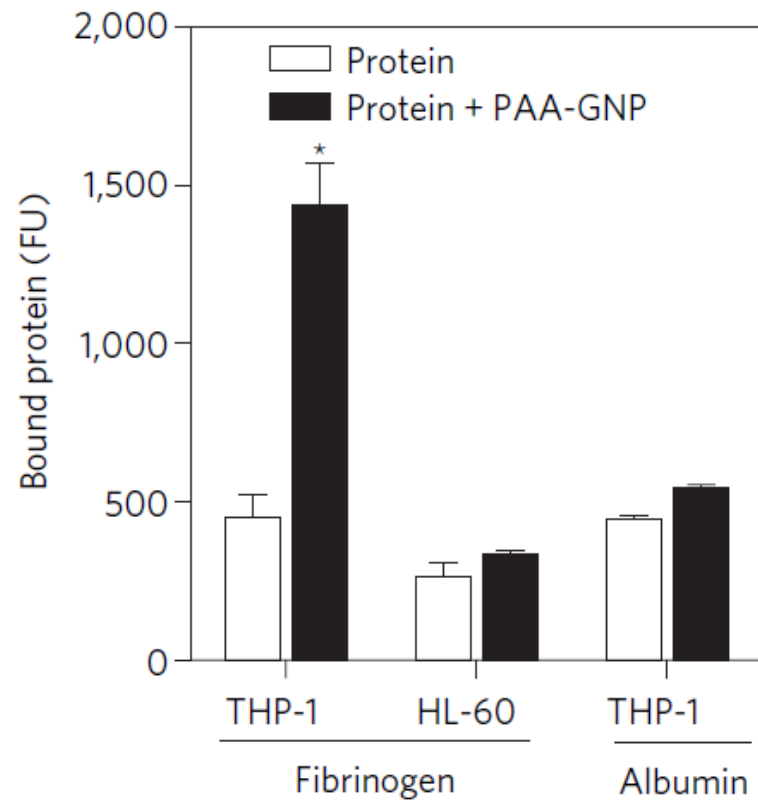


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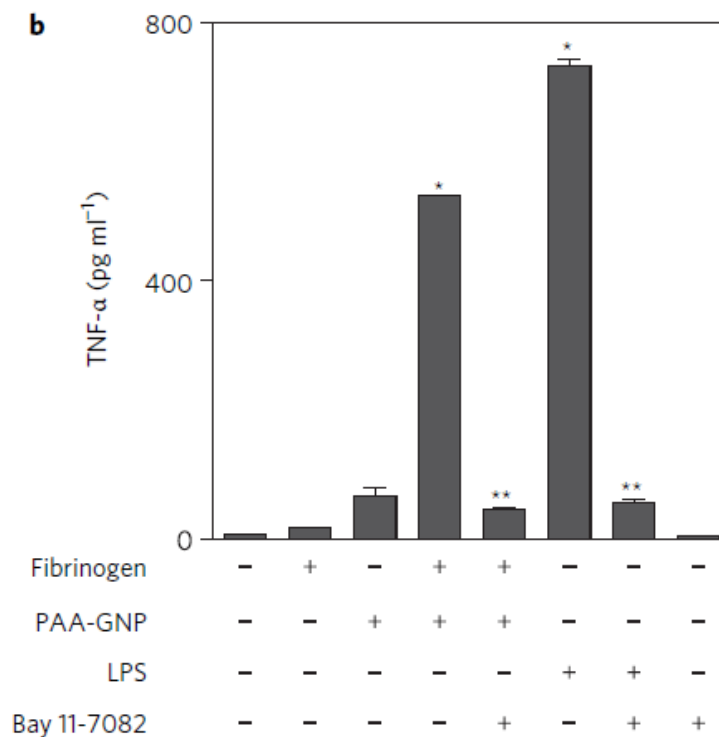
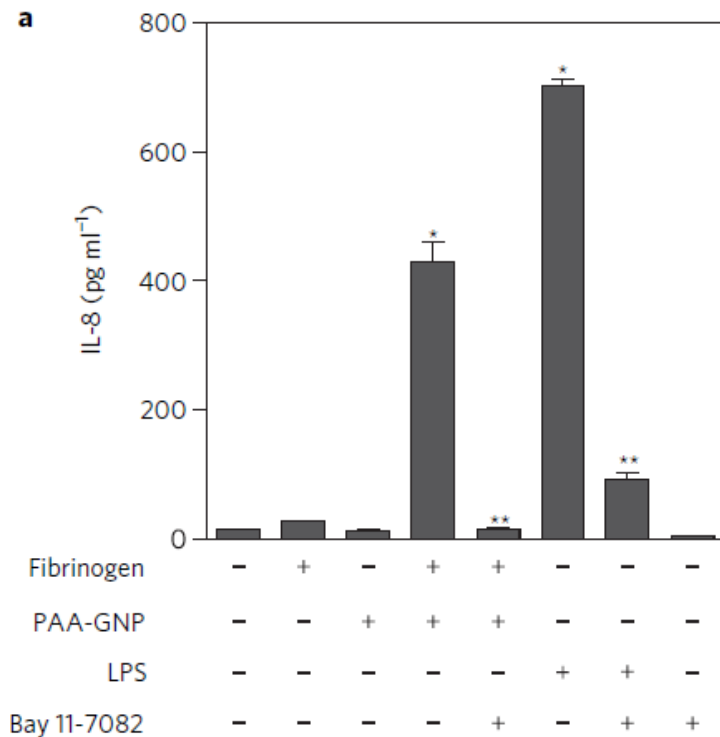
- SDS-PAGE of human plasma proteins bound to PAA-GNP with diameters of 5, 10 and 20 nm, three protein bands were observed at 65, 55 and 45 kDa.

The binding capacity of the nanoparticles for the purified human fibrinogen was determined by ultracentrifugation.

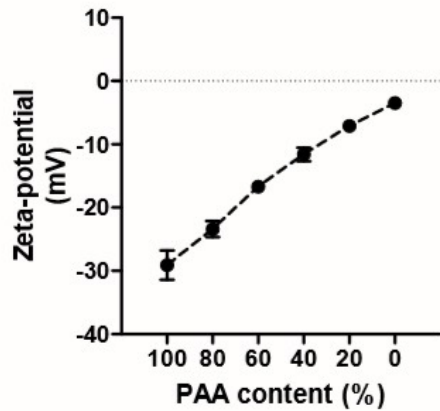
- 20 nm PAA-GNP bound significantly less fibrinogen than the 5 nm particles on a mass basis. But, when adjusted for total surface area, the binding was found to be similar
- The addition of PAA-GNP resulted in a loss of protein secondary structure, as indicated by the progressive increase in ellipticity



- Experiments done for 20nm GNP did not show any significant bound protein due to lack of exposure of γ C terminus

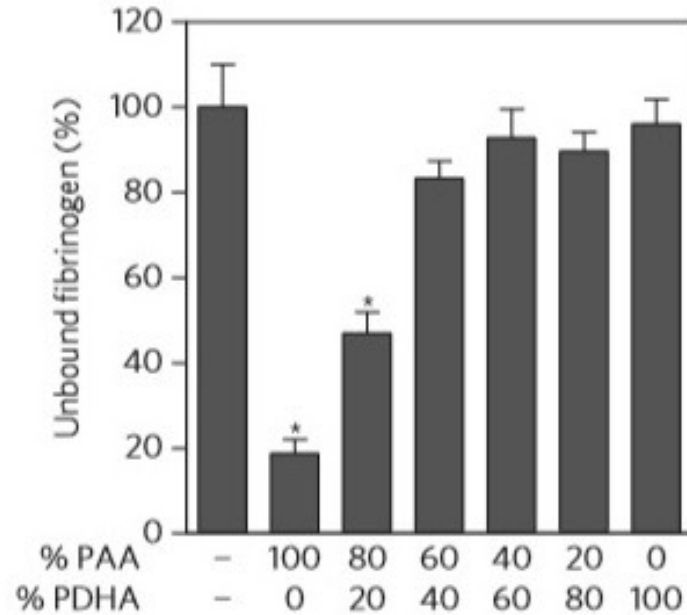


- As a result of NF-κB signalling, inflammatory cytokines are released by the cells.
- THP-1 cells were subjected to GNP-fibrinogen, non conjugated GNPs and fibrinogen. BAY 11-7082 which is an inhibitor of NF-κB pathway was used as a negative control and LPS treated cells were used as a positive control.



Zeta-potentials of PAA-PDHA-GNPs

Neutral polymer: Poly(2,3-hydroxyl-propylacrylamide) (PDHA)



Fibrinogen binding decreased significantly with 20% PDHA and was completely inhibited at higher proportions of PDHA.

Tissue Engineering

Tissue engineering is the use of a combination of cells, engineering and materials methods, and suitable biochemical and physicochemical factors to improve or replace biological tissues



Controversial image

- In August 1997, Joseph Vacanti's group published the ground-breaking paper
- Ear (made from cow cartilage) was grafted onto the mouse back
- The technology has improved since then including use of human cells



Tissue Engineering

- Generating whole tissue in-vitro and replacing damaged tissues in patients
- Generating part tissue in-vitro and replacing damaged portions of the tissues in patients
 - Putting in scaffolds to regenerate damaged part of the tissue in-vivo
- Adding support/filler material to support the function of damaged tissue

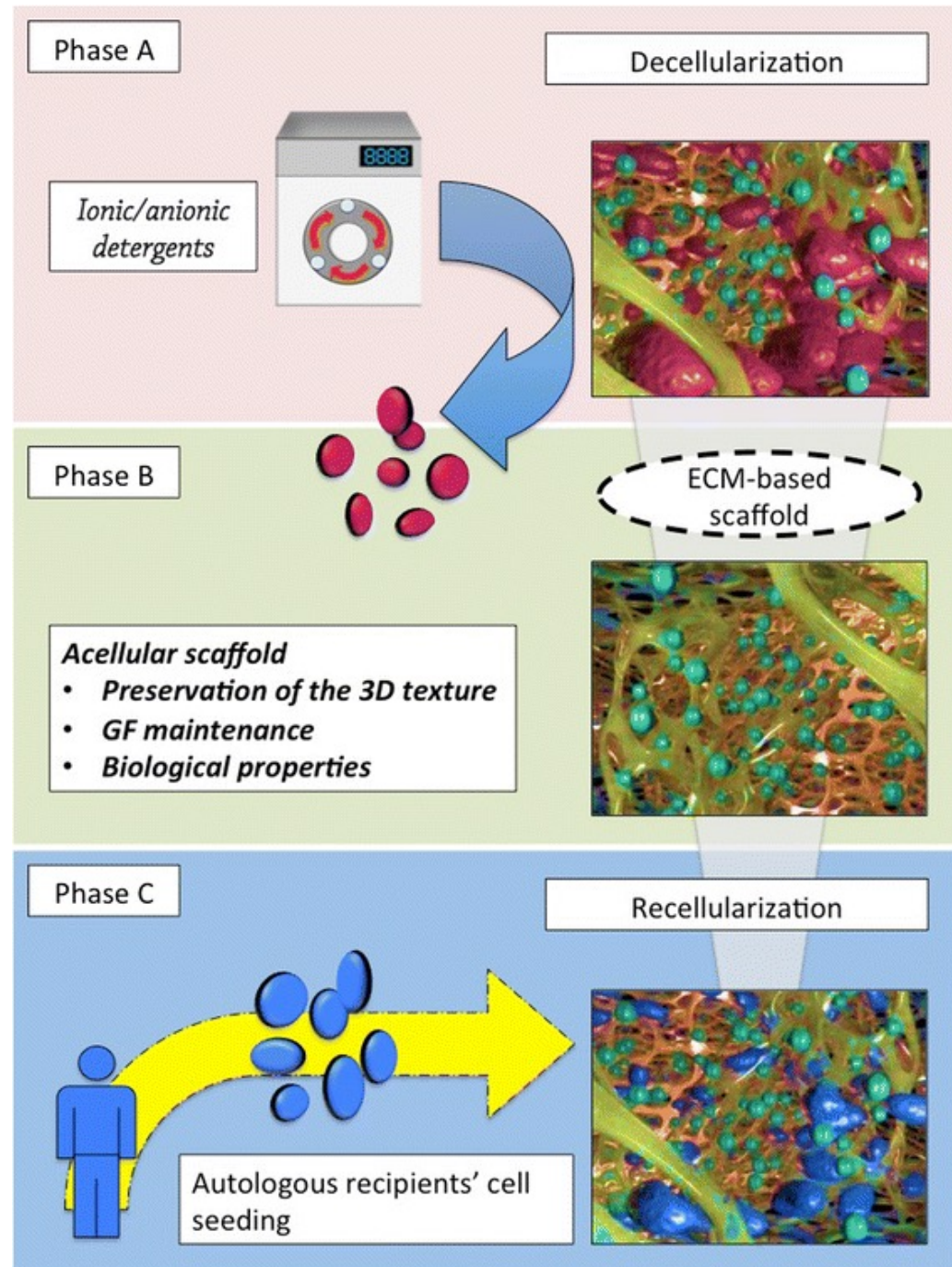
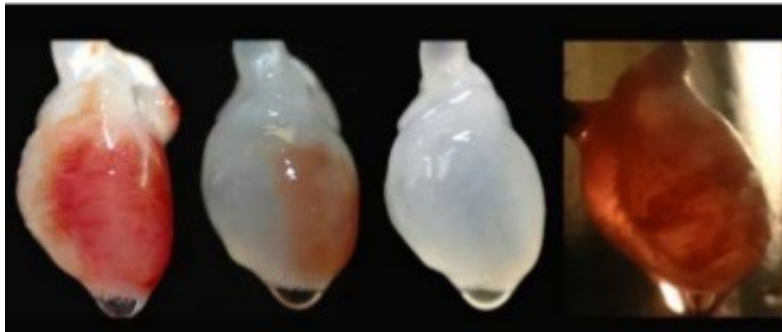
Whole tissue replacement

- Transplant from another human is the most clinically used practice
 - Lack of donors
 - Spread of disease
 - Immune rejection: half life of organs is around 10-15 years
- Transplant from genetically modified pigs has gained attention
 - However, immune rejection is still a major issue

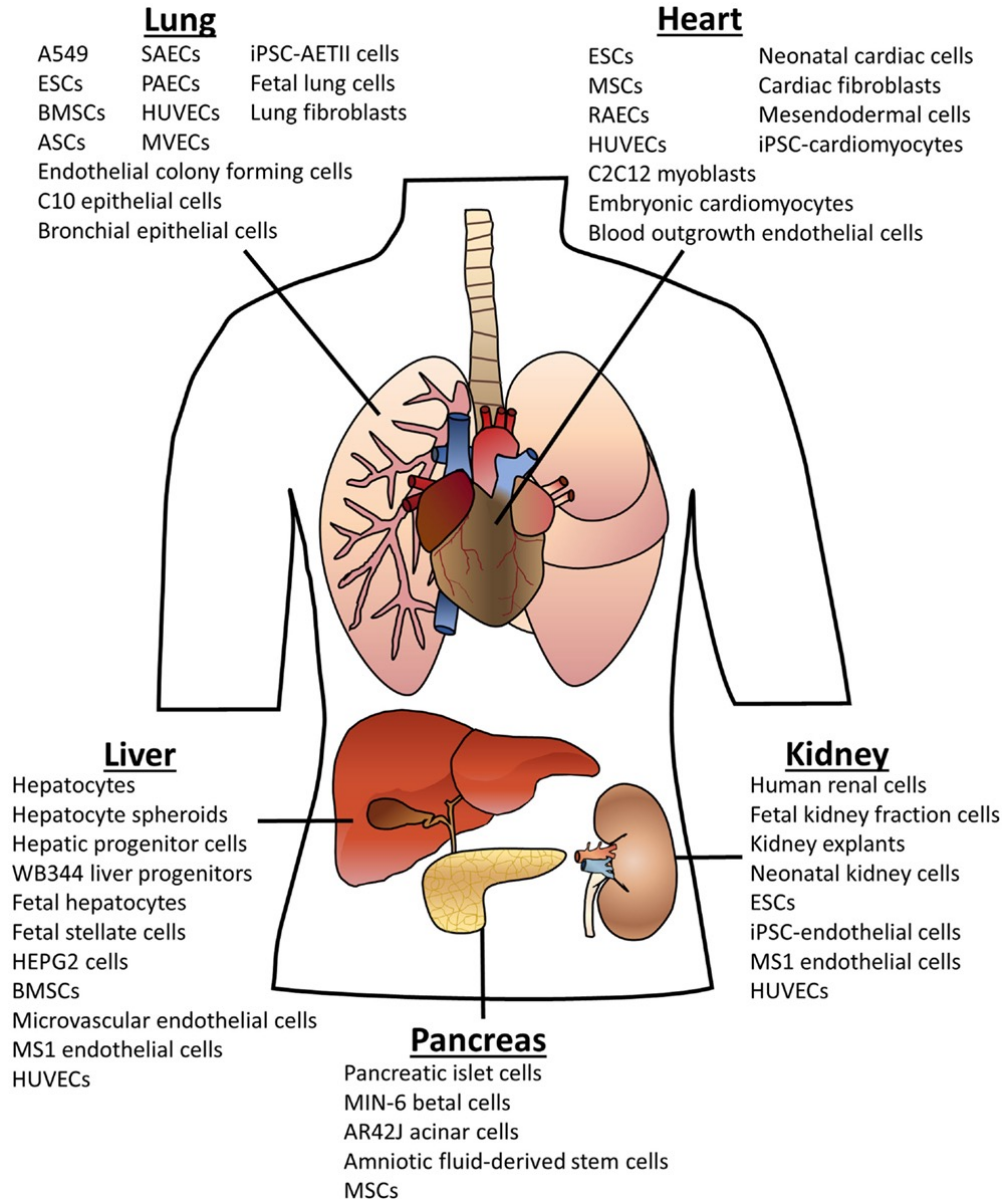
Whole tissue replacement

- Not many bioengineering approaches are available
- Major limitation is lack of blood vessels and hence the tissues are thin
- 3D printing of tissues has gained momentum but is still nascent
- Most widely used is decellularization of tissues from cadavers and animals

- Remove cells via detergents
- Seed autologous cells
- Implant in the body



Recellularization of scaffolds



Recellularization with primary cell is important

Difficult to ascertain the fate of these implanted cells

- Another approach is not recellularize at all
- Implant just the decellularized scaffold
- Rely on ability of primary cells and stem cells to repopulate the tissue
- One limitation is that small capillaries don't survive this process



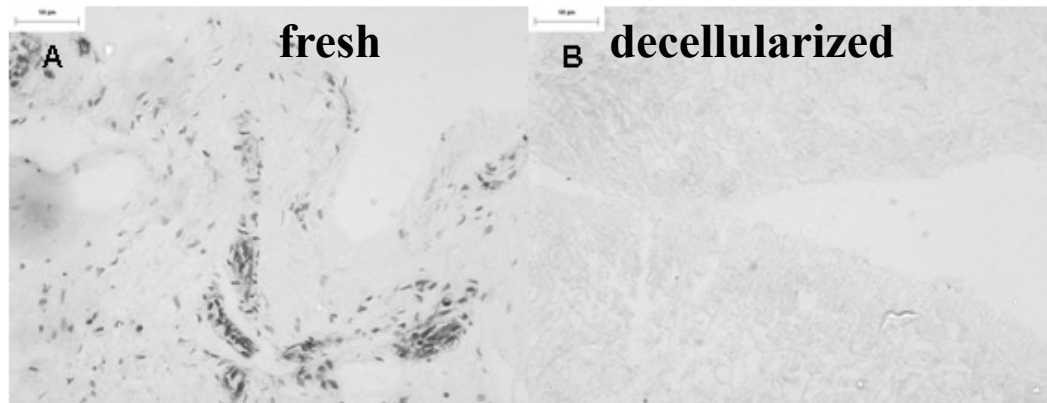
Clinical Applications:
dCELL® Vascular Patch &
Dermapure for Advanced Woundcare

Slide acknowledgements: Prof. Shervanthi Homer-Vanniasinkam

Decellularization: Considerations

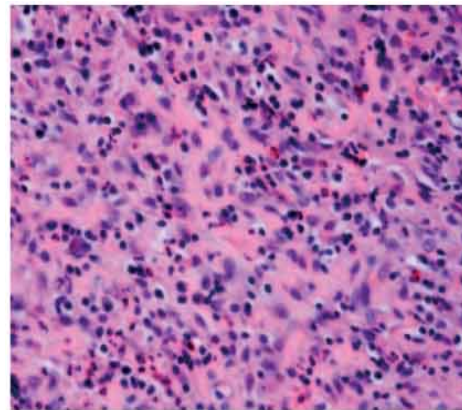
- Mammalian cells have surface antigens called alpha-gal
- If grafts are derived from non-human source then these antigens can produce strong immune response
- Important to check the absence of alpha-gal from decellularized implant

alpha-gal immunoperoxidase stain

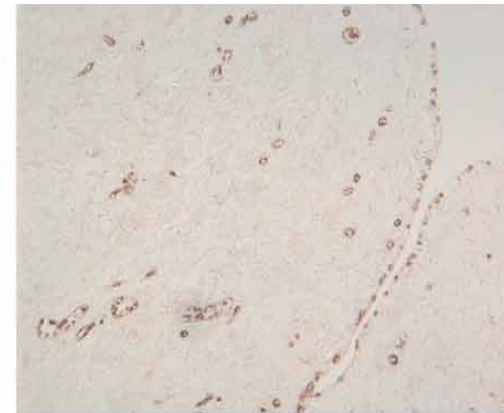


Complication following TE graft implantation: hyperacute rejection

Spark et al. *Br J Surg* 2008; 95: 582-585



b Haematoxylin and eosin



c Immunostain for α -Gal

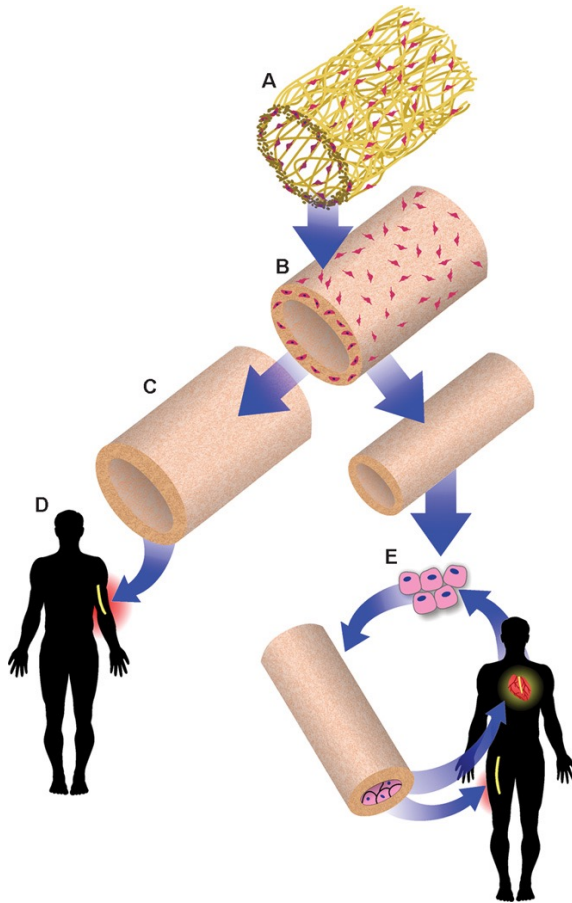
Vascular graft to bypass blockage

Slide acknowledgements: Prof. Shervanthi Homer-Vanniasinkam

Tissue engineered vascular graft (TEVG)



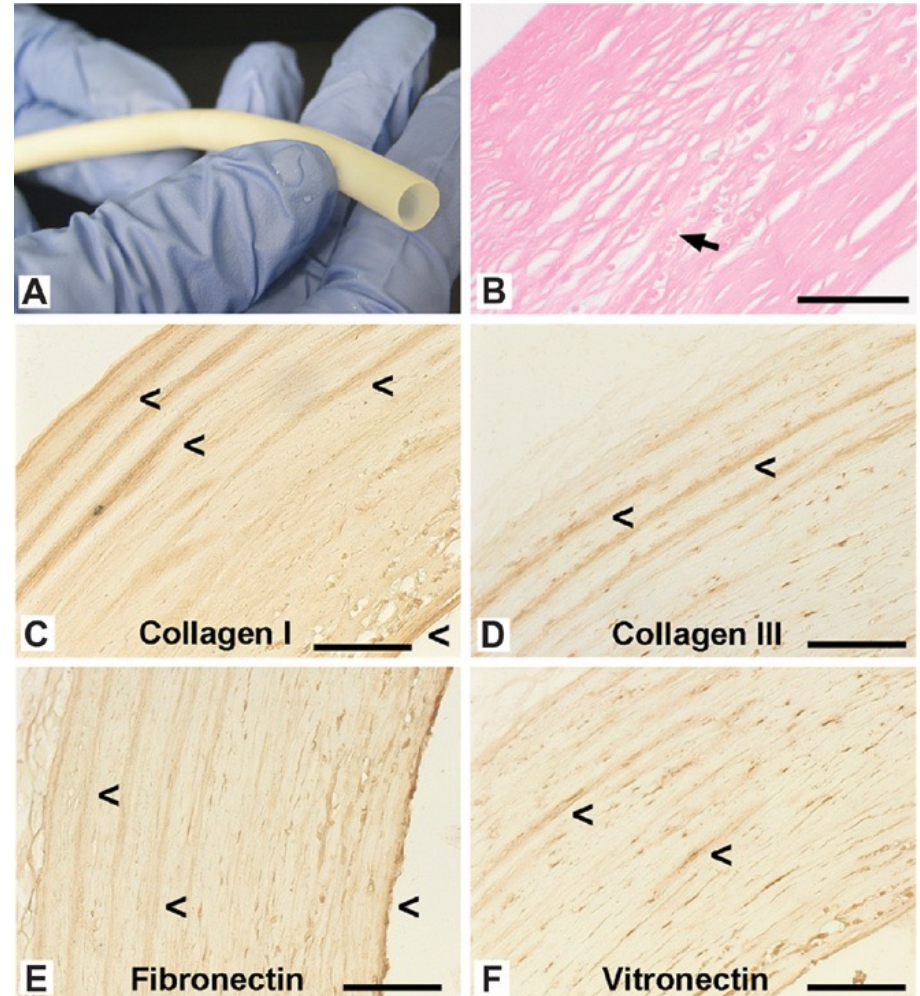
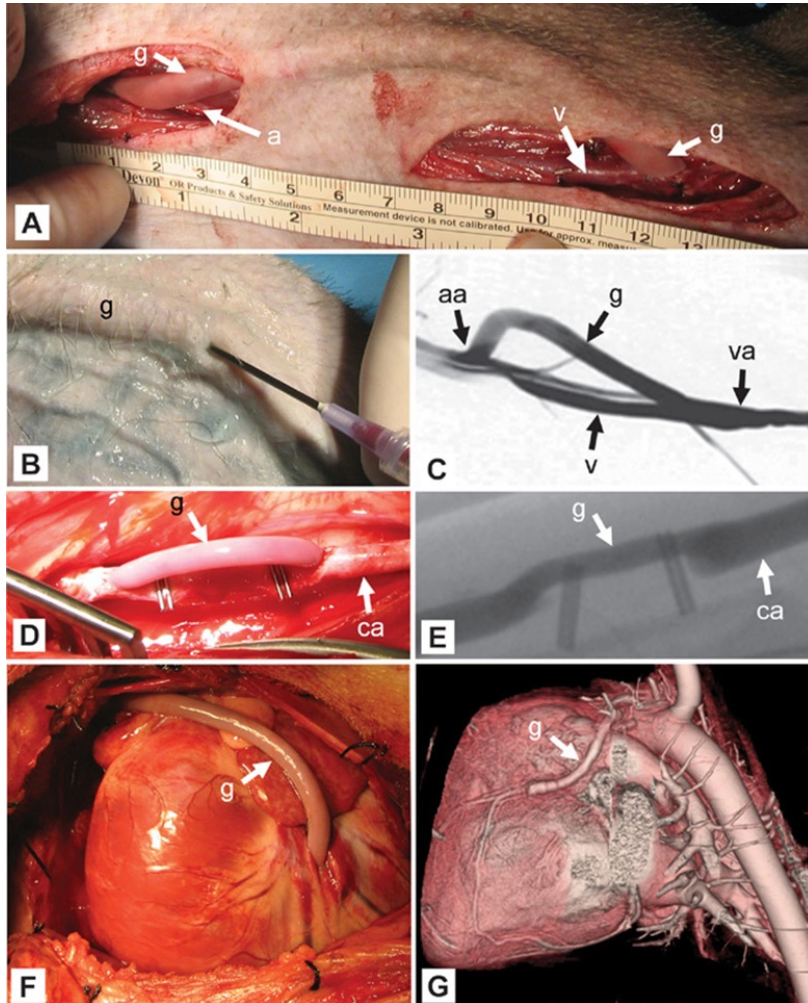
HUMACYTE



- Seed human cells on platform (A)
- Cells produce ECM (B)
- Cells decellularized and implant ready to in human (C)
- Shown to recellularize better in humans

Slide acknowledgements: Prof. Shervanthi Homer-Vanniasinkam

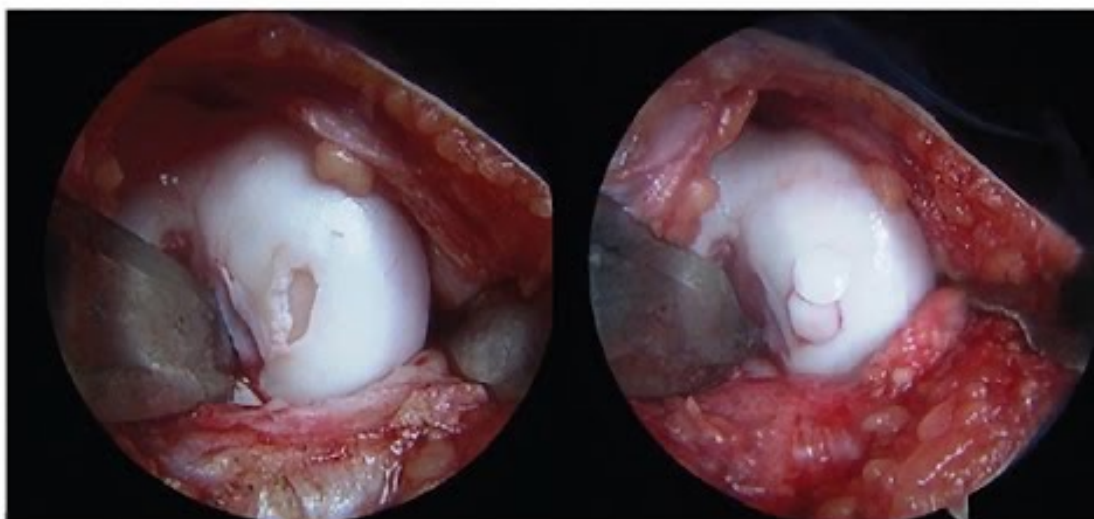
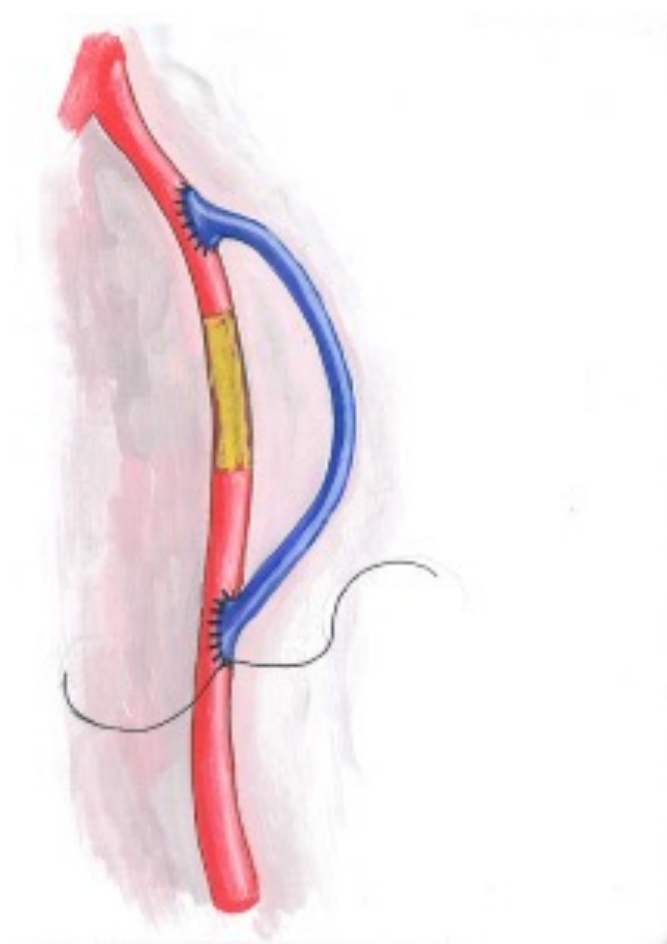
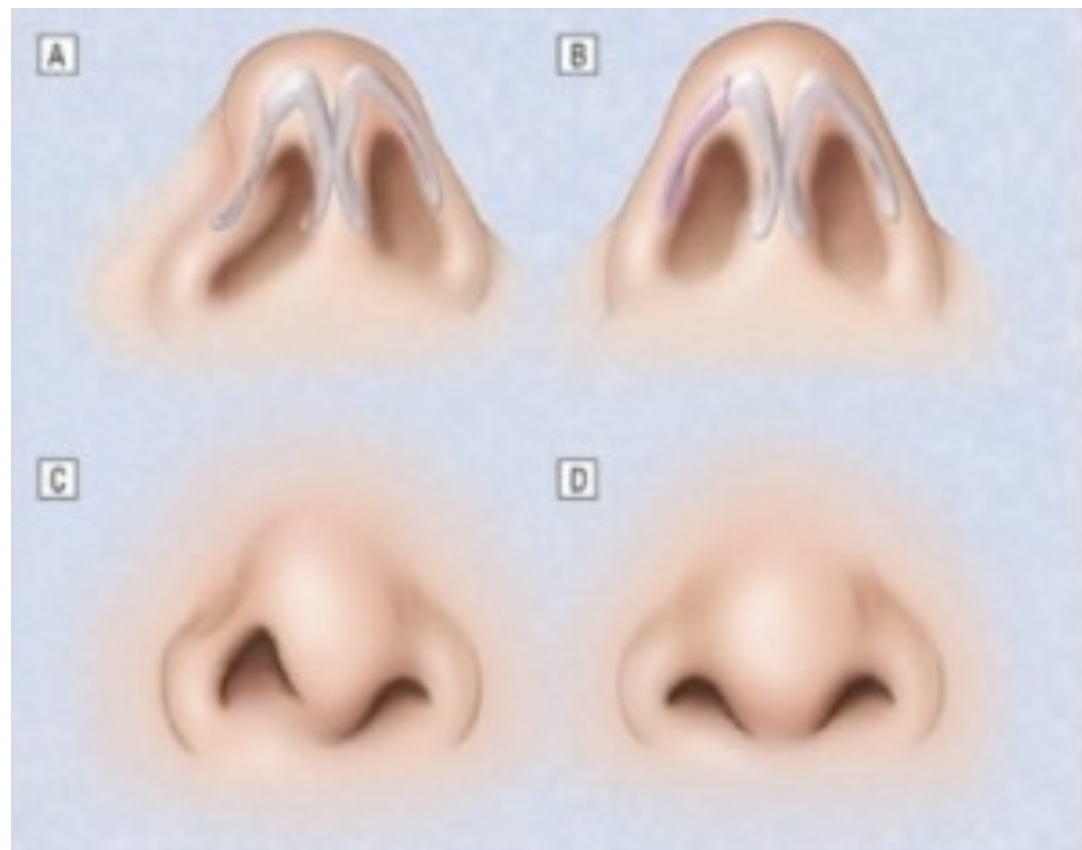
Humacyte used for bypassing blocked vessels



Slide acknowledgements: Prof. Shervanthi Homer-Vanniasinkam

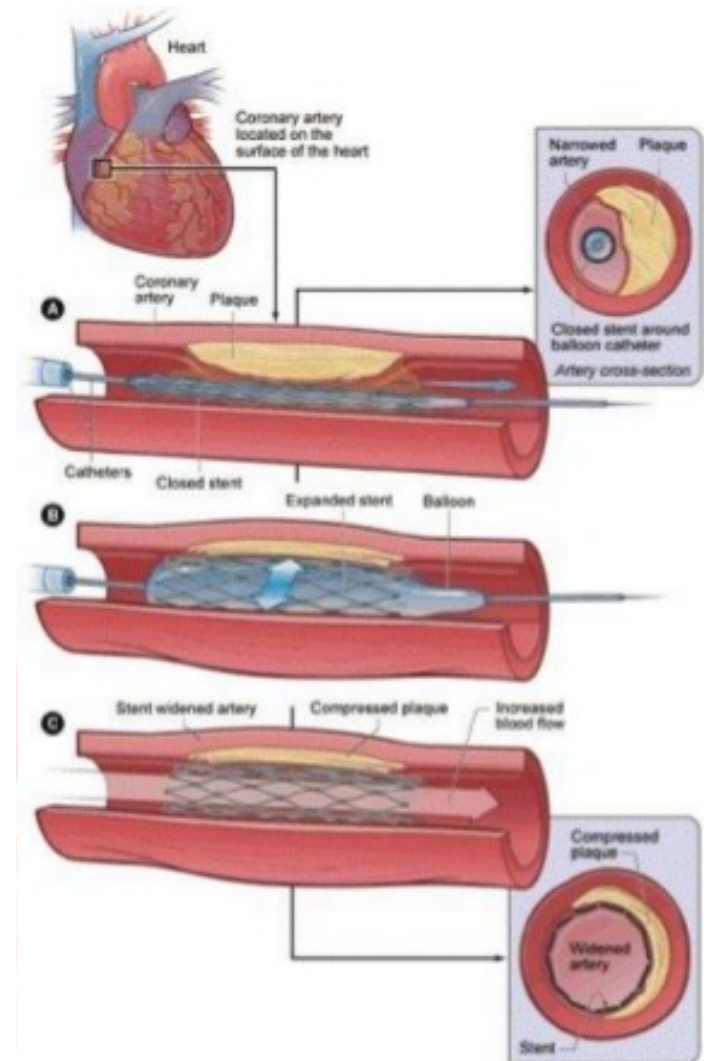
Tissue Engineering – regenerating part of a tissue

- Much more successful
- Two approaches
 - Material + cells in-vitro followed by implantation
 - Material implantation for scaffolding
- Ultimate goal is to get the tissue back to its native state
- Applications:
 - Cartilage defects
 - Vascular grafts
 - Heart patch



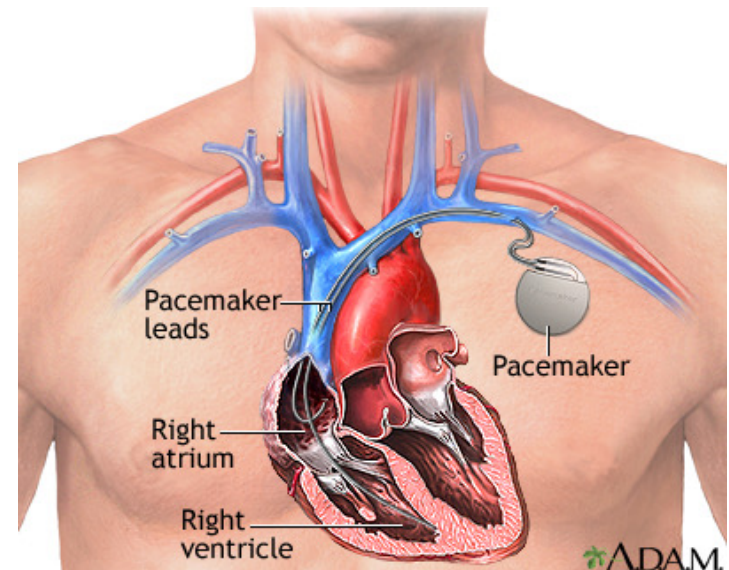
Tissue Engineering – Support function of damaged tissue

- Mainly to provide support/filler material for proper function of damaged tissue
- Stents: Remove occlusions from vessels



Tissue Engineering – Support function of damaged tissue

- Dental and Orthopaedic metal implants
 - Knee
 - Long bones
- Pacemaker
- Contact lens: Widely used



Drug Delivery in Tissue Engineering

- Depending on the application most of the these implants would need to release chemicals and drugs for effective functioning
 - VEGF and PDGF for blood vessel formation
 - Antibiotic for prevention of infections
 - Immunosuppressants for acceptance of graft
 - Adsorbed proteins and growth factors
 - Other small molecules

Review

Polymeric Growth Factor Delivery Strategies for Tissue Engineering

Ruth R. Chen¹ and David J. Mooney¹⁻⁴

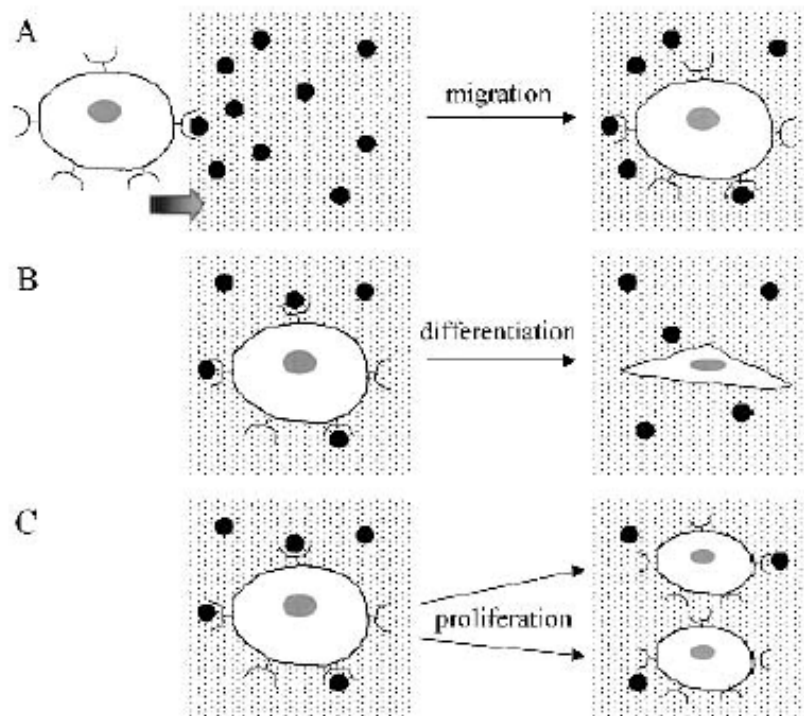
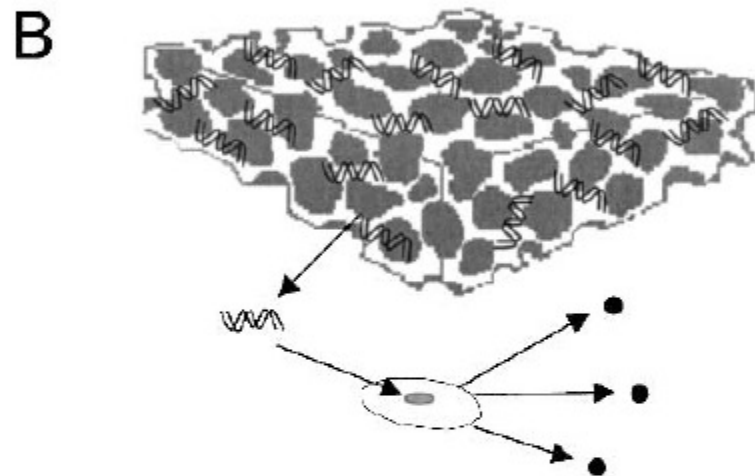
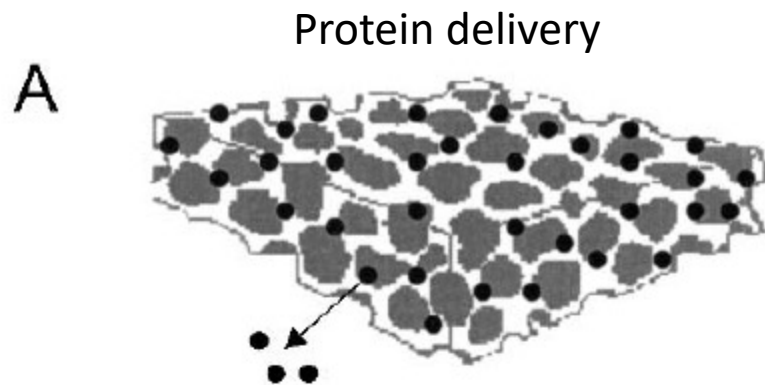


Fig. 1. Growth factor binding to a cell surface receptor can elicit different types of responses to regenerate tissue in a defect site. A, Chemotactic responses to growth factor binding result in cell migration into a defect site. B, Cell differentiation can occur to provide the needed cell population. C, Proliferation of cells within the site fill the tissue defect.

Table I. A List of Growth Factors Commonly Used in Tissue Engineering

Growth factor	Abbreviation	Molecular weight (kDa)	Known activities
Epidermal growth factor	EGF	6.2	Proliferation of epithelial, mesenchymal, and fibroblast cells
Platelet-derived growth factor	PDGF-AA	28.5	Proliferation and chemoattractant agent for smooth muscle cells; extracellular matrix synthesis and deposition
	PDGF-AB	25.5	
	PDGF-BB	24.3	
Transforming growth factor- α	TGF- α	5.5	Migration and proliferation of keratinocytes; extracellular matrix synthesis and deposition
Transforming growth factor- β	TGF- β	25.0	Proliferation and differentiation of bone forming cells; chemoattractant for fibroblasts
Bone morphogenetic protein	BMP-2	26.0	Differentiation and migration of bone forming cells
	BMP-7	31.5	
Basic fibroblast growth factor	bFGF/FGF-2	17.2	Proliferation of fibroblasts and initiation of angiogenesis
Vascular endothelial growth factor	VEGF ₁₆₅	38.2	Migration, proliferation, and survival of endothelial cells

rH, recombinant human.



DNA delivery

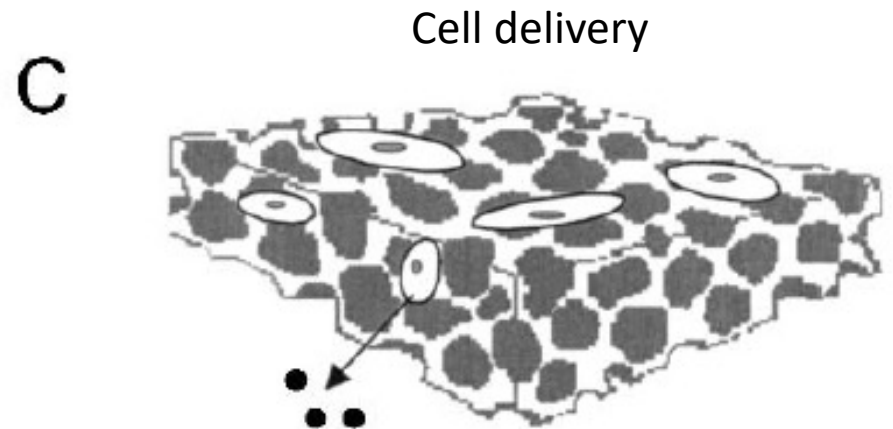
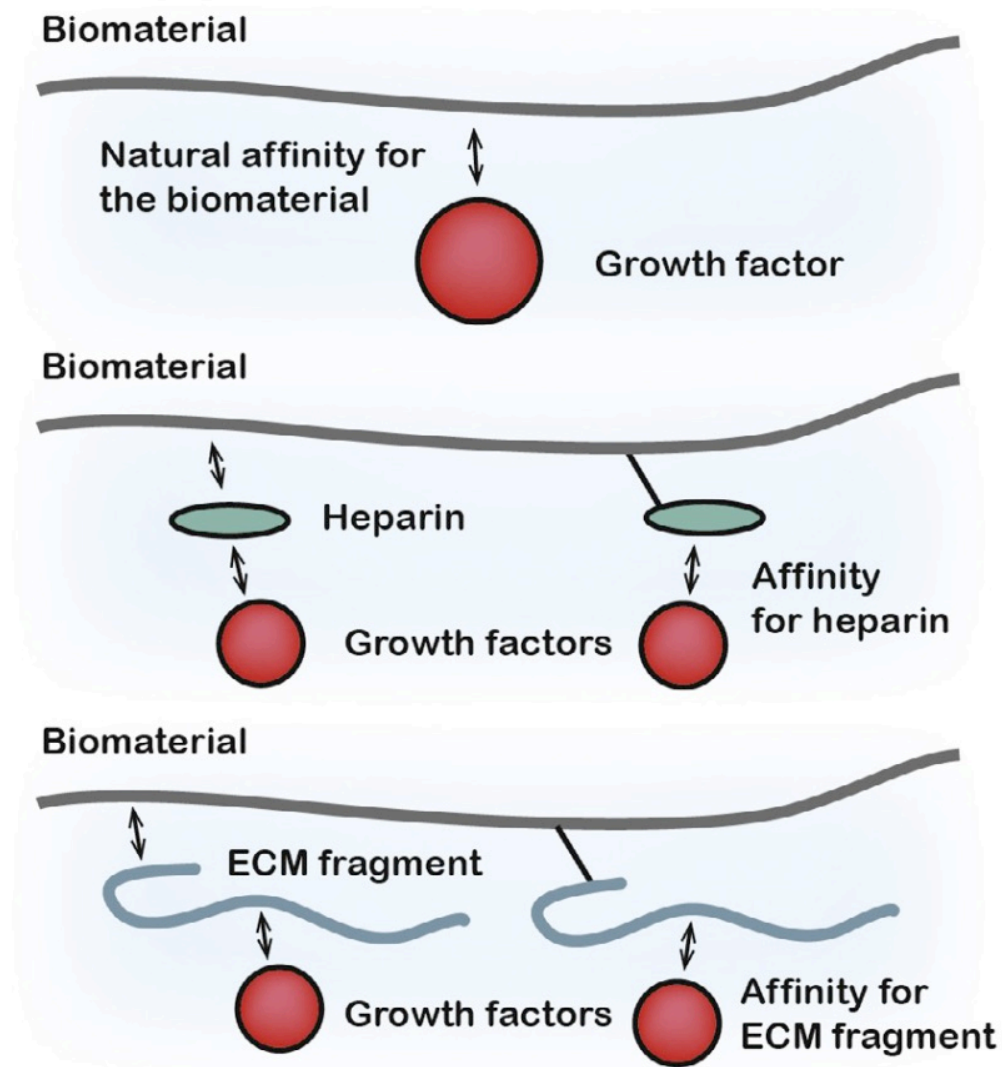
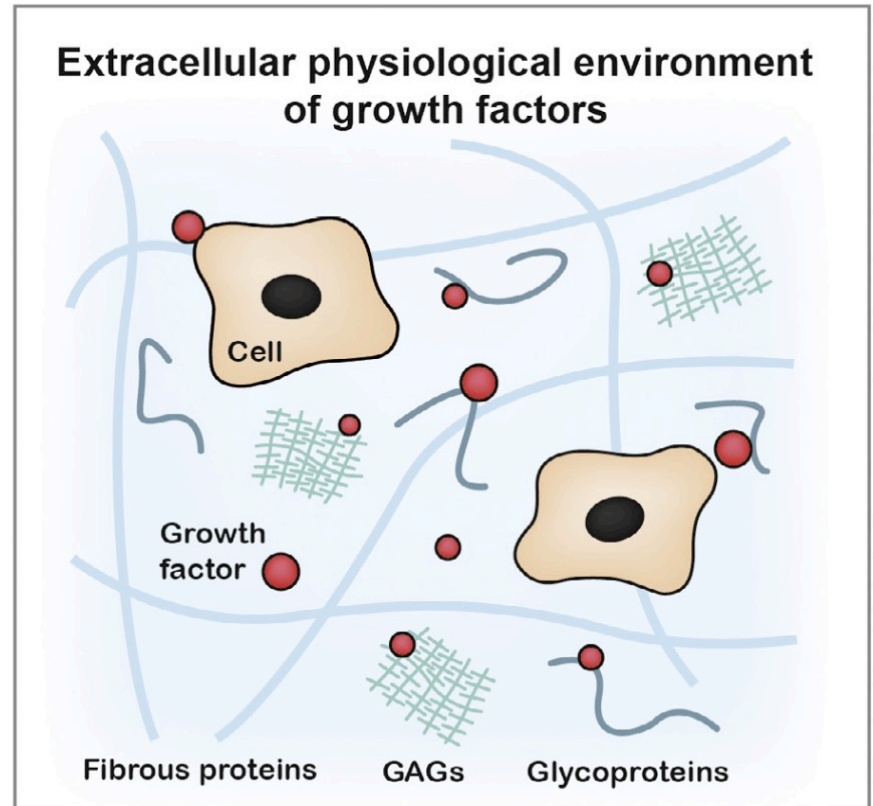
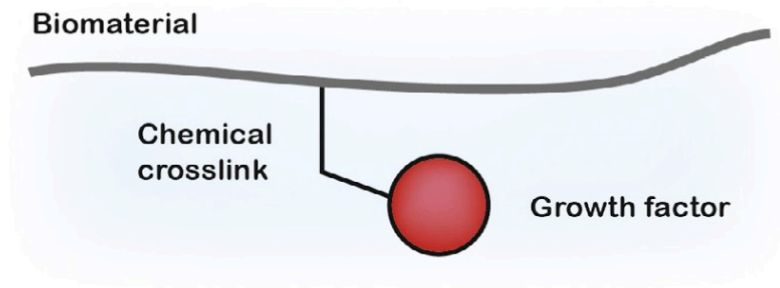


Fig. 3. Three primary polymeric growth factor delivery strategies. A, Growth factors are embedded within the polymer matrix and released. B, Genes encoding a growth factor are embedded within the polymer matrix and released, followed by cellular uptake and expression of the gene to produce growth factor. C, Growth factor is released from cells seeded on the polymer matrix that secrete the factor.

Non-covalent affinity for biomaterial matrices

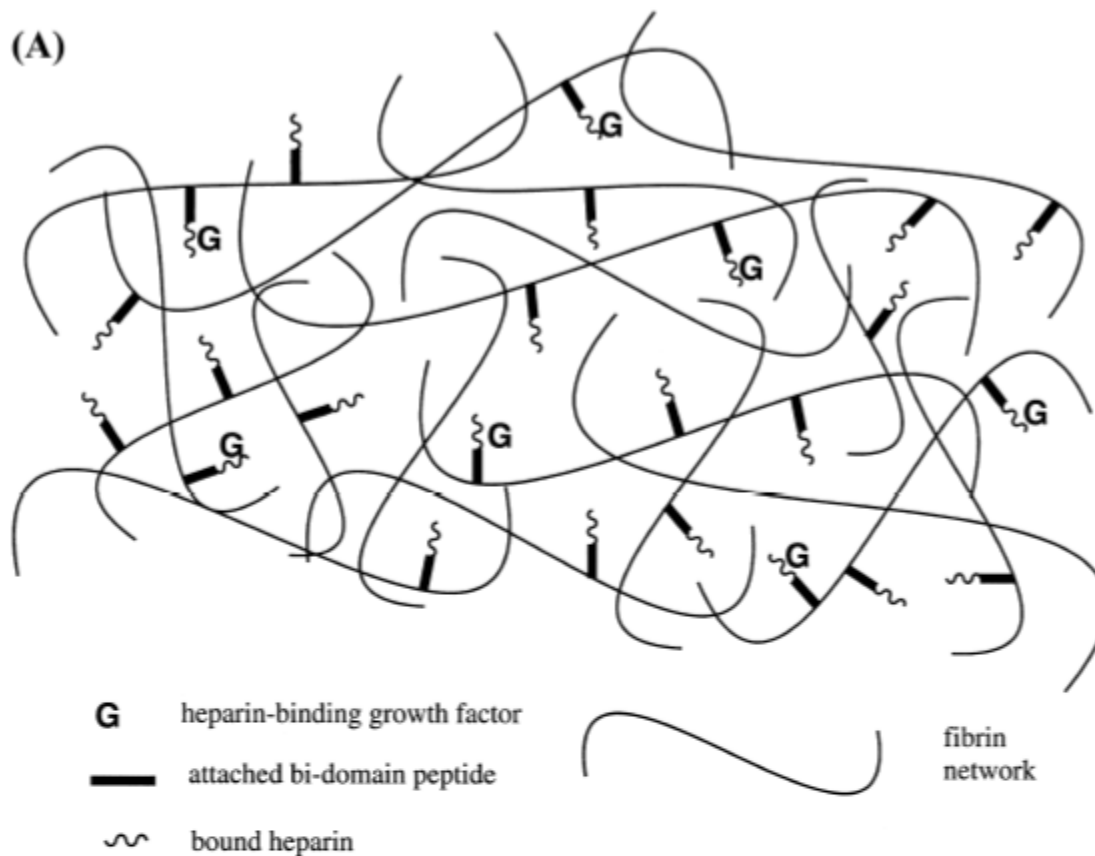


Covalent attachment to biomaterial matrices



Development of fibrin derivatives for controlled release of heparin-binding growth factors

Shelly E. Sakiyama-Elbert^{a,b}, Jeffrey A. Hubbell^{a,*}



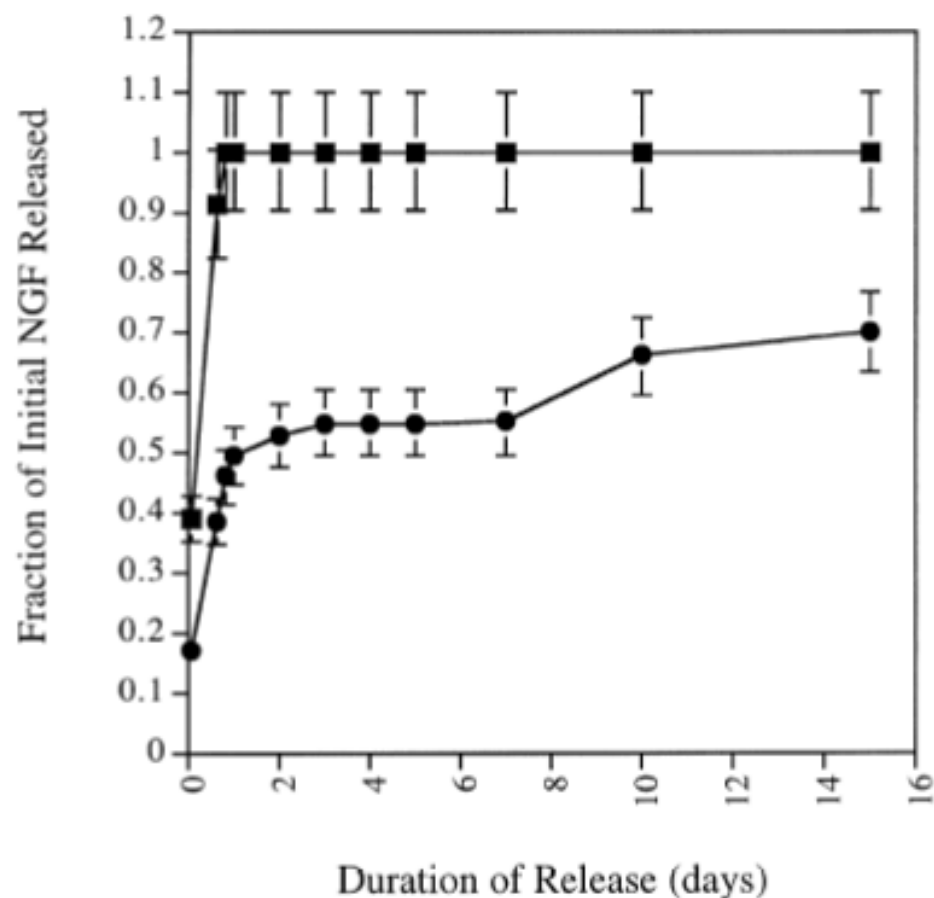


Fig. 2. Release profile of β -NGF diffusion from fibrin matrices in the presence or absence of the heparin-containing delivery system. (■) Unmodified fibrin with 100 ng/ml of β -NGF in the culture medium. (●) Heparin-containing matrices with 100 ng/ml of matrix-interacting β -NGF. Mean values and standard deviation are shown. These results demonstrate that despite β -NGF's weak heparin-binding affinity, the release of β -NGF from fibrin matrices is prolonged by a heparin-containing drug delivery system.

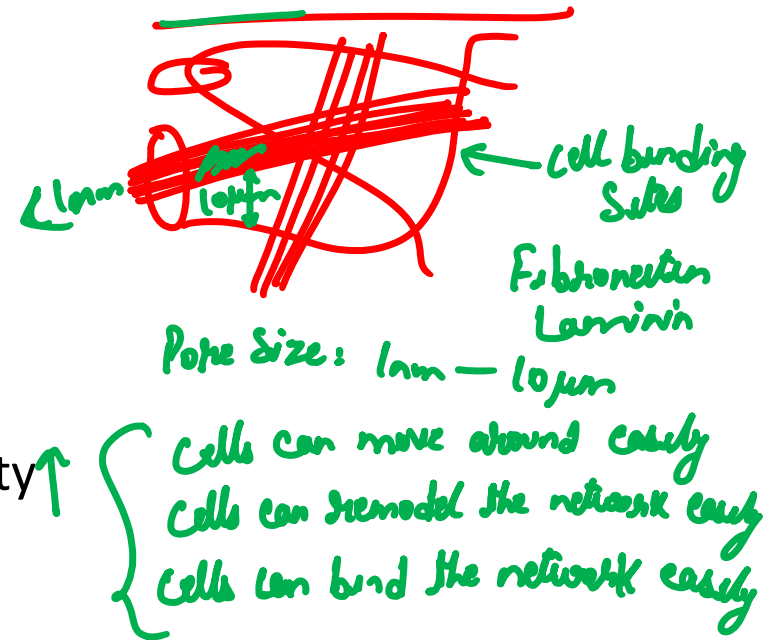
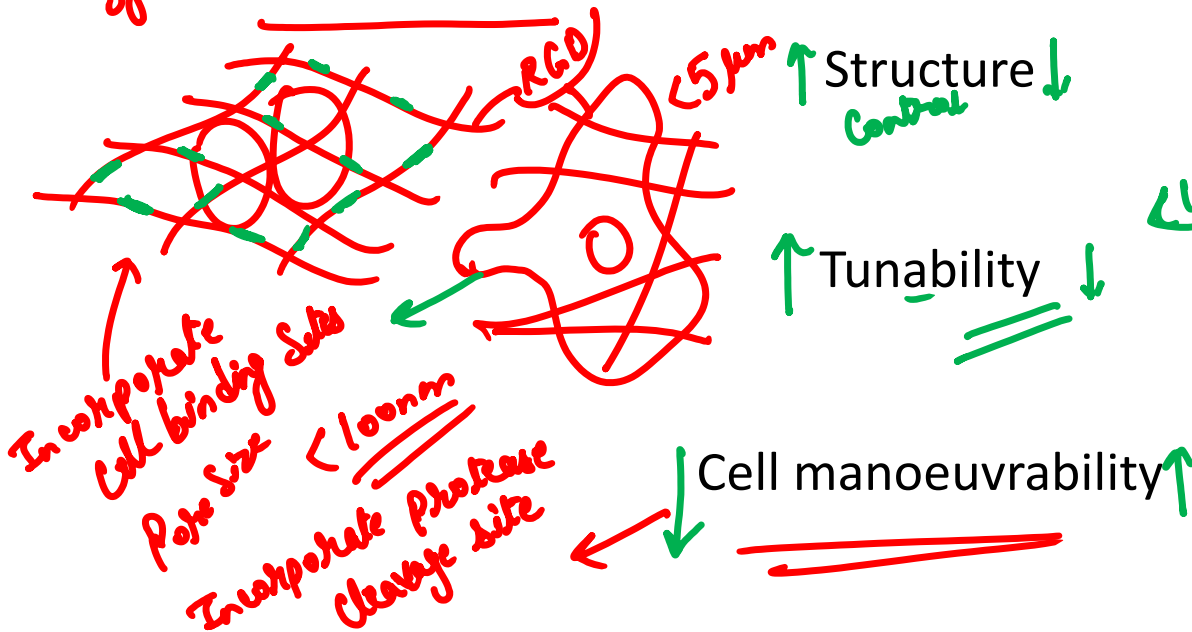
Natural hydrogels vs synthetic hydrogels for tissue engineering

PEG hydrogels

Collagen hydrogels

Synthetic material

Natural material

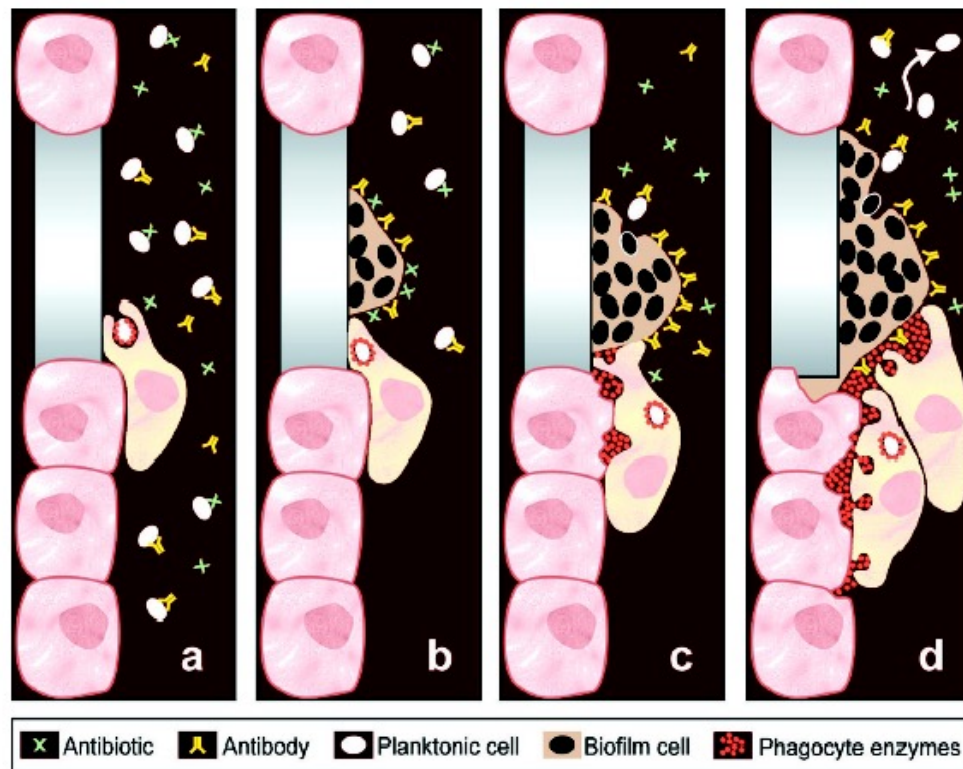


Implant Associated Infections

Definitions

- **Biofouling**
 - The phenomenon of protein, cellular and bacterial (microorganism) attachment to material surfaces that leads to loss of device function, infection and implant failure
- **Biofilms**
 - Microorganisms (bacteria) in colonies on surfaces form layers (2-100 organism thick) which are composed of organism, cellular material, extracellular polysaccharides, environmental adsorbates and debris. This surface composite is called biofilm or slime

Fig. 1. Diagram of a medical biofilm. (A) Planktonic bacteria can be cleared by antibodies and phagocytes, and are susceptible to antibiotics. (B) Adherent bacterial cells form biofilms preferentially on inert surfaces, and these sessile communities are resistant to antibodies, phagocytes, and antibiotics. (C) Phagocytes are attracted to the biofilms. Phagocytosis is frustrated but phagocytic enzymes are released. (D) Phagocytic enzymes damage tissue around the biofilm, and planktonic bacteria are released from the biofilm. Release may cause dissemination and acute infection in neighboring tissue.



Controlled release of antibiotics from plasma-deposited membranes

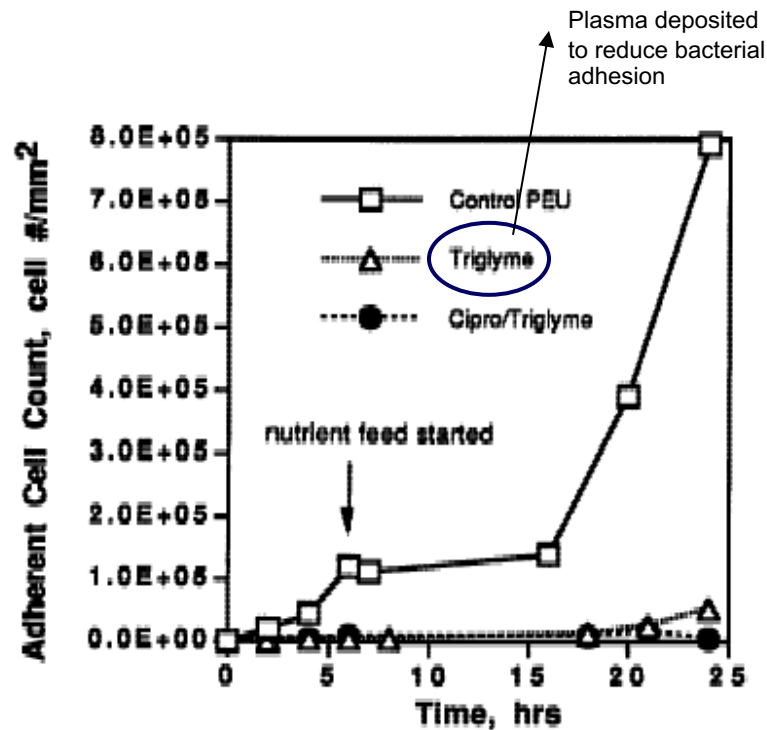


Figure 5. Adherent cell numbers per unit area with elapsed time for three different materials: PEU control, PEU with triglyme only, and PEU with pBMA barrier membrane releasing ciprofloxacinTM. Total experiment is shown.